INDIA LEARNING ACADEMY

Warehousing: Management & Design

17th Sep'13

Warehouse – Few facts …

1. Initiated as Port side establishments to facilitate trade
   - First recorded history in India = 1612, Surat
2. Various names for same purpose (broadly): Godowns, depots, warehouse….
3. Logistics spend constitutes 13% of Indian GDP
   - 20% of this is on account of Warehousing = USD 47 Bn (2012-13)
4. Management models: Outsourced (60%); In-house (40%)
   - 70% of Outsourced warehouses managed by C&F; 30% by 3 PL
   - 3 PL industry growing by 25% YoY (Frost & Sullivan)
5. Warehousing sector to grow at 35% YoY over next 5 years (Cushman & Wakefield)

Warehouse: What it is ?

















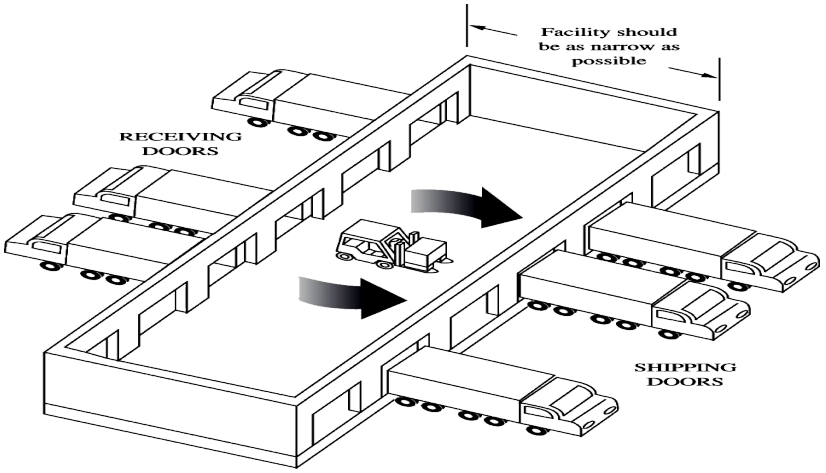
Warehouse: What it is not !!

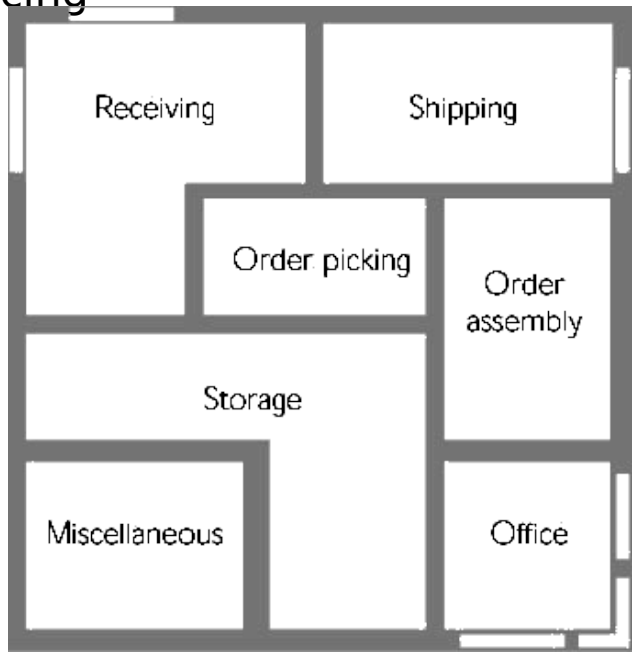


Warehouse: Definition

Warehousing refers to the activities involving storage of goods on a large-scale in a systematic and orderly manner and making them available conveniently when needed.

- Referred as transportation at zero miles per hour
- Consists of:
  - Storage area: Good stock, Quarantine stock; Aisles for movement
  - Office area: Order management and invoicing
  - Docks for loading & unloading





Need for Warehouse?

1. To Consolidate Product:

- _Amortization of logistics fixed cost by maximising load
- _Reduction of transportation cost by converting LTLs into FTLs

2. To realize economies of scale

- Bulk buying from vendors for better prices, hence heavy storage
- Large batch size manufacturing to amortize high set up cost

3. To provide value added Processing

- Assembling or customization required as close as to customers
- Labelling, kitting or packaging change required

4. To reduce Response time

- Quicker delivery to customer needs by being close to market place
- High stocks needed to cope up with peak of demands like seasonality
- Skewness of demand and sales forces to response with stock pile up in remaining time

Types Of Warehouse:

1. **Retail Distribution System:**
- Supplies to retail stores directly
- Regular and scheduled delivery to multiple stores with multiple items
- Like Walmart, Home town

2. **Service Part Distribution center:**
- Holds service parts for capital equipments like automobiles, airplane, computer services
- High inventory of large no of spare parts, challenging facility
- Very unpredictable demand and quick service required with least lead time

3. **E- commerce distribution center:**
- Receives small orders from individual by email ,Fax
- Unpredictable demand and fast delivery required

4. **3 PL or shared distribution system:**
- Outsource WH activity, mutliple customers from single facility
- Gaining economies of scale
- Can be an overflow facility for surge in demands

Pre Requisites for Warehouse operations

1. Licenses
   - Shops & Establishment/MCD license
   - Gram Panchayat NOC
   - VAT registration license
   - Weight and measurement license

2. Lease agreement & Management (3PL) agreement
3. CST Number
4. TIN number
5. Safety compliance: Fire extinguisher; earthing of electrical wiring; Safety shoes; Safety caps; Driving license for forklift
6. Material master configuration
7. Customer master configuration
8. Optional:
   - Import Export Code
   - Re packing license
   - DG installation

Warehouse Activities:

1. Receive goods:
   - Accepts goods from Outside transportation or attached factory & accepts responsibility
   - Check the goods against an order & the bill of loading
2. Identify and check the goods :
   - Items are identified with the appropriate stock-keeping unit (SKU) number (part number) & the quantity received recorded
3. Dispatch goods to storage (Put Away)
   - Goods are sorted & put away
4. Hold goods :
   - Goods are kept in storage & under proper protection until needed

Warehouse Activities:

5. Pick goods :

- Items required from stock must be selected from storage & brought to a marshalling area

6. Marshal shipment

- Goods making up a single order are brought together & checked for omissions or errors; order records are updated

7. Dispatch shipment:

- Orders are packaged, shipping documents are prepared, & goods loaded on the vehicle

8. Operate an information system

- A record must be maintained for each item in stock showing the quantity on hand, quantity received, quantity issued, & location in the warehouse

Warehousing Cost

- Constituents of warehousing costs:
  - Warehouse lease (~ 40-60% of total expense)
  - Capital Deployed (~ 15%)
    - Storage & materials handling equipment
    - IT & Network
  - Manpower Employed (~20%)
  - Miscellaneous (~5%)
    - Electricity
    - Consumables

- Warehouse design hugely influences cost of managing a warehouse; namely:
  - Space:
    - Type of storage: Horizontal / Vertical
    - Warehouse equipments
    - Product characteristics
  - Stock Flow: Number of manpower employed

- Other factors impacting costs:
  - Inventory levels
  - Site selection

Warehouse: Site Selection

Warehouse site selection

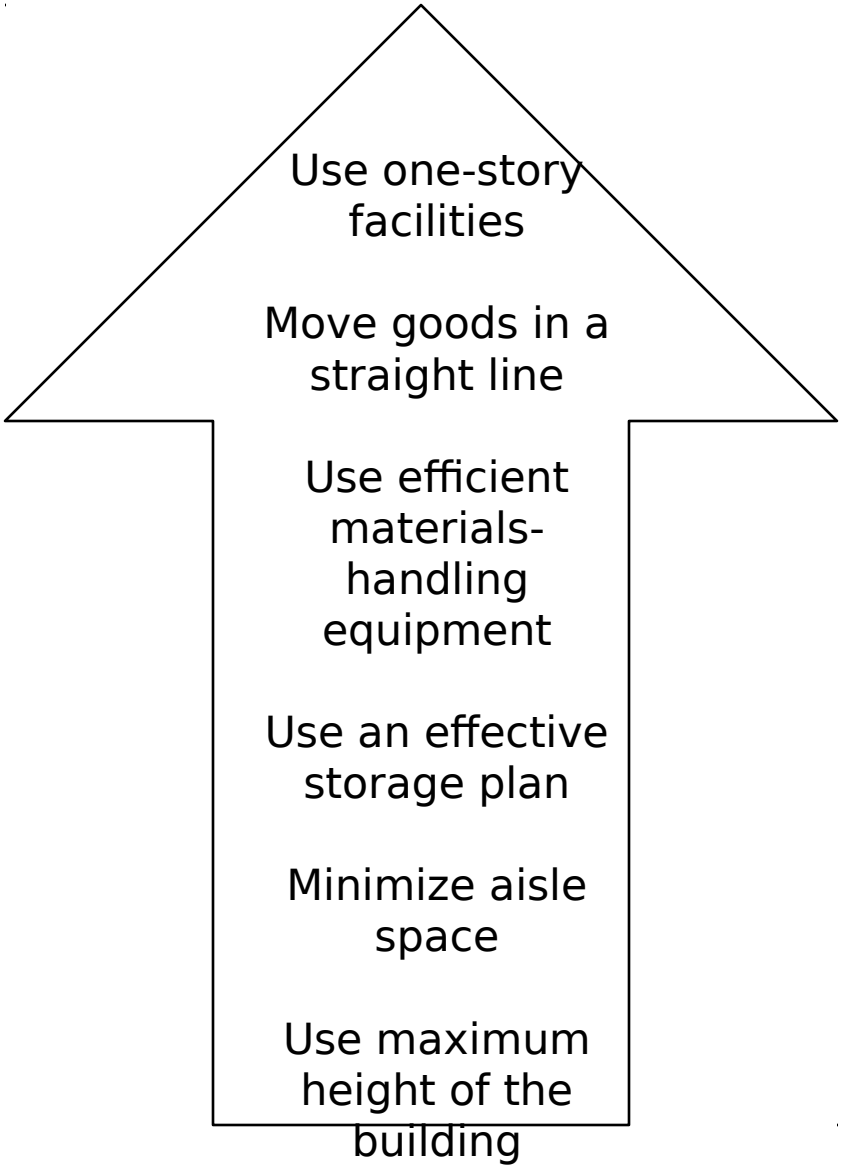
The considerations of warehouse site selection revolve around two major factors;

1. Service
2. Cost

The other factors affecting site selection are:

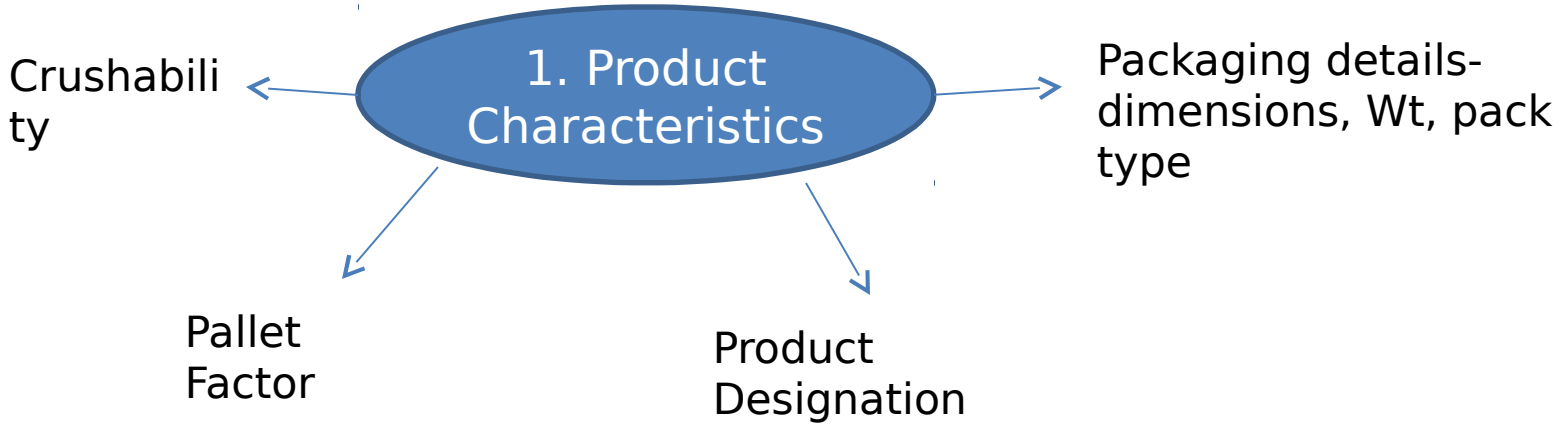
a. Infrastructure
b. Market
c. Access
d. Transportation cost
e. Availability
f. Product
g. Regulations
h. Local levies

Principles of Warehouse Design

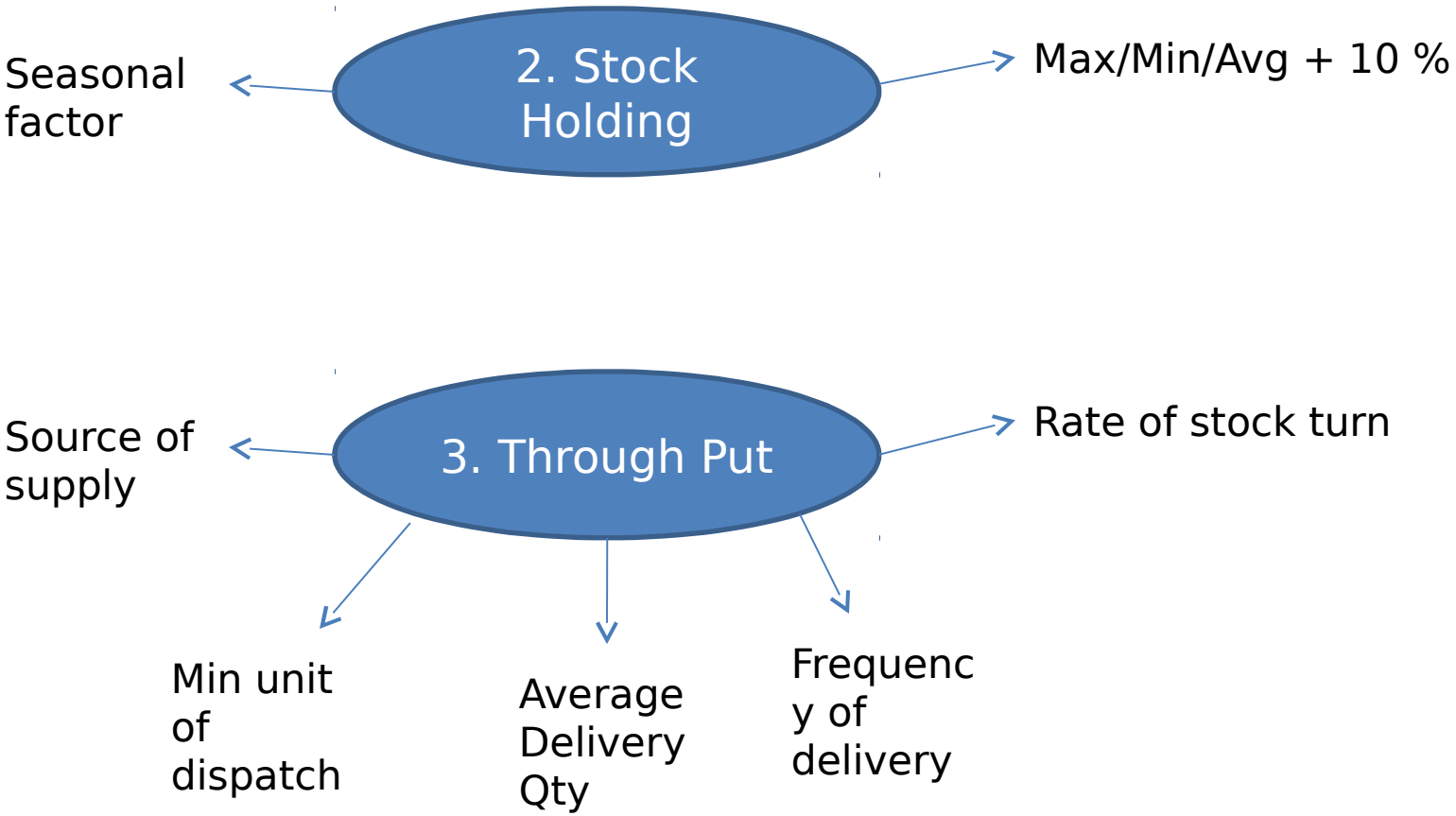


Basic data for WH Design:

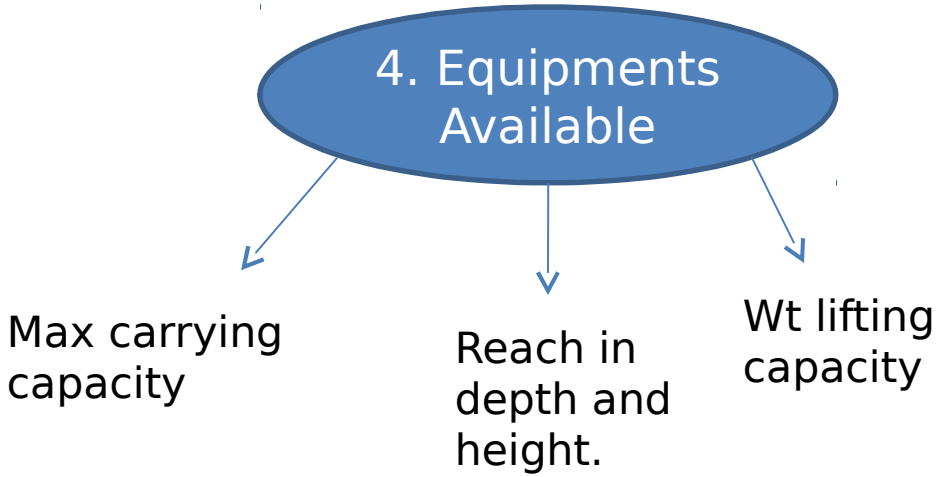
- Essential information to be collected for the design of any warehouse,

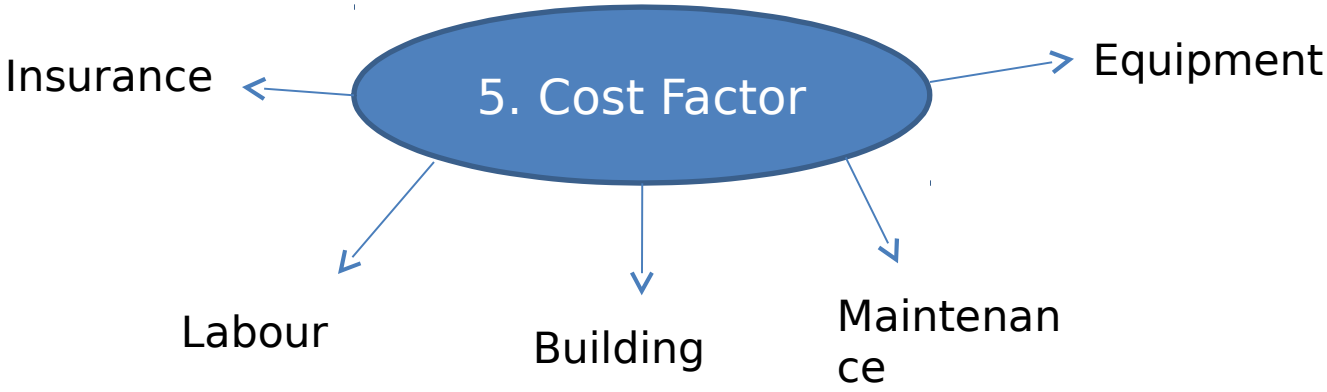


Check List of Basic data



Check List of Basic data





Warehouse Design Planning

Divided in two parts:

1. Storage space layout design

2 . Movement flow design of people and goods

A. Storage Space Layout

Two Types of Storage:

1. Block storage



- Floor storage, multiple pallets of same SKU stored together

- Feasible in scenario of narrow SKU range, high stock holding per SKU

- 20 % SKU contributing 80% of sale ideal for block storage

- Min equipment usage, easy to pick
- Limited gang way space

Two Types of Block storage -

a. Fixed location type : *(Max Inventory level taken for calculation)*

- SKU assigned a permanent location, & no other items are stored there
- Fixed-location systems usually have poor cube utilization
- Usually used in small warehouses; throughput is small, & there are few SKUs

b. Random by location *: (Avg+10% Inventory level taken for space calc)*

- Goods stored wherever there is appropriate space
- Advantage is improved cube utilization
- It requires accurate and up-to-date information
- Warehouses using floating-location systems are usually computer / WMS based

Block Storage Pallet Area Calculation:

- Calculation of No of pallet rows for Fixed type block storage:

- Input data - a. No of pallets in a Stack height

b. Row length in no of pallets

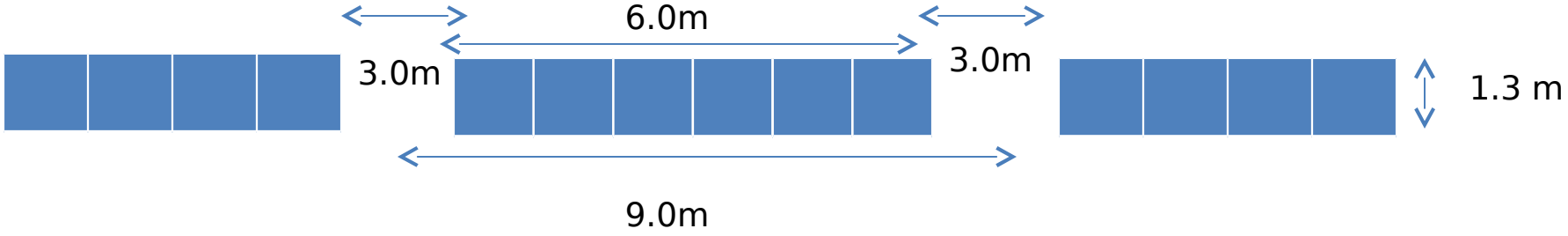
c. No of pallets in a row= a*b

d. Being block storage , space planning as per max stock level of each SKU. e. Divide Max stock (d) by no of no pallets in a row (b)

* Additional 1 row of pallets to be added per 3 row of pallets for stock rotation

$$\text{No of rows required} = \frac{\text{Maximum stock}}{\text{No of pallets in a row}} + 1 \text{ additional row (Where grtr than 3)}$$

- The utilization should be 80% of WH super built up space..



- Area occupied by 18 pallets (assuming 3 high) plus aisle is 9 m* 1.3 m = 11.7 sq.m
- Area occupied by 1 pallet @ 100% utilization = 11.7 / 18 = 0.65 sq. M
- Add 30 % loss for utilization and 10 % loss for transverse gang / aisleways
- Which comes to 0.91 sq.m per pallet area

A (contd...). Storage Space Layout

2. Individual Access system:



Single Rack System

- Racking system with access to each individual pallets
- Ideal for high SKU range with more order lines to pick
- Better utilization of space
- Thumb rule: With 5 pallets high 50 % more storage space created of floor area used
- More space required in gang/aisle ways
- Equipments required to pick

Types of Racking System

1. Single deep rack :



2. Double deep Rack :



3. Push Back rack :



4. Drive In or Drive through Rack :

